

E-Commerce Customer Behavior Analysis – Data Analytics Case Study

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Tools Used

Google Sheets, Google BigQuery, Tableau

1. Project Overview

This project analyzes customer demographic and behavioral data from an e-commerce platform.

The purpose of the analysis is to understand how customer characteristics such as age, gender, location, and income influences engagement with the platform.

By analyzing browsing behavior and time spent on the website, the project aims to uncover patterns that can help businesses design better marketing strategies and improve customer targeting.

2. Business Problem

E-commerce businesses collect large amounts of customer data but often fail to convert it into meaningful insights.

Understanding customer behavior is essential for:

- Improving marketing strategies.
- Increasing customer engagement.
- Identifying high-value customer segments.

This project focuses on transforming raw customer data into actionable insights.

3. Business Questions

1. What is the distribution of customers by gender?
2. Which age groups represent the largest customer segments?
3. Which customer segments have the highest income levels?
4. Which users spend the most time on the website?
5. Which locations contribute the highest number of customers?

4. Dataset Description

The dataset used in this project contains demographic and behavioral information about customers .

of an e-commerce platform. The dataset was obtained from Kaggle.

Dataset columns include:

- Customer ID
- Age
- Gender
- Location
- Annual Income
- Purchase History
- Browsing History
- Product Reviews
- Time on Site

For better analysis, an additional variable Age Group was created during the data preparation stage.

5. Tools Used

- Google Sheets – Data cleaning and preparation.
- Google BigQuery – SQL data analysis.
- Tableau – Data visualization and dashboard creation.

6. Data Cleaning

Before starting the analysis, several data cleaning steps were performed.

Cleaning steps included:

- Removing duplicate records
- Checking for missing values
- Standardizing gender categories
- Verifying numeric data formats
- Creating an Age Group column for segmentation

Age groups used:

- 18–25
- 26–35
- 36–45
- 46–60

7. SQL Analysis

Example SQL query used to analyze gender distribution:

```
SELECT Gender, COUNT(*) AS customer_count  
  
FROM `casestudy-490307.Ecommerce.customer_info`  
  
GROUP BY Gender  
  
ORDER BY customer_count DESC;
```

The screenshot shows the Google Cloud BigQuery console interface. On the left, a sidebar displays the project hierarchy: 'casestudy-490307' > 'Ecommerce' > 'customer_info'. The main area shows an 'Untitled query' with the following SQL code:

```
1 select Gender, COUNT(*) AS customer_count  
2 FROM `casestudy-490307.Ecommerce.customer_info`  
3 GROUP BY Gender  
4 ORDER BY customer_count DESC;
```

Below the query editor, a status bar indicates 'Query completed' and 'Using on-demand processing quota'. The 'Query results' section is active, displaying a table with 3 rows and 2 columns: 'Gender' and 'customer_count'.

Row	Gender	customer_count
1	Female	26
2	Male	18
3	Other	6

On the right, a 'Reference' panel shows the schema for the 'customer_info' table, listing fields like Customer ID, Age, Gender, Location, Annual Income, Purchase History, Browsing History, Product Reviews, Time on Site, and Age Group with their respective data types.

Example SQL query used to analyze average income by age group:

```
SELECT `Age Group`, AVG(`Annual Income`) AS avg_income  
  
FROM `casestudy-490307.Ecommerce.customer_info`  
  
GROUP BY `Age Group`  
  
ORDER BY avg_income DESC;
```

("Here the `int()` function is used because the average income value is returned as a float (for example, 59.0). The `int()` function converts this float value into an integer (59), which makes the result cleaner and easier to display.")

The screenshot displays the Google Cloud BigQuery console interface. The top navigation bar includes the Google Cloud logo and a search bar. The main content area is divided into three sections: a left sidebar with a project tree, a central query editor, and a right sidebar with a reference table.

Query Editor: The central pane shows an SQL query titled "Untitled query". The query is as follows:

```
1 SELECT `Age Group`, cast( AVG( `Annual Income` ) as int64 ) as avg_income
2 FROM `casestudy-490307.Ecommerce.customer_info`
3 GROUP BY `Age Group`
4 ORDER BY avg_income DESC;
5
```

Below the query editor, a status bar indicates "Query completed" and "Using on-demand processing quota".

Query results: The bottom section displays the results of the query. It includes a table with the following data:

Row	Age Group	avg_income
1	46-60	82200
2	36-45	73500
3	26-35	52579
4	18-25	43750

Reference: The right sidebar shows a reference table for "customer_info". It lists the table's location as "casestudy-490307.Ecommerce.customer_info" and provides details such as "50 rows - 17.89 KB - Last modified at 15 Mar 2026, 12:32". A table of field names and types is also displayed:

Field name	Type
Customer ID	INTEGER
Age	INTEGER
Gender	STRING
Location	STRING
Annual Income	INTEGER
Purchase History	STRING
Browsing History	STRING
Product Reviews	STRING
Time on Site	FLOAT
Age Group	STRING

("Here the `int()` function is used because the average income value is returned as a float (for example, 59.0). The `int()` function converts this float value into an integer (59), which makes the result cleaner and easier to display.")

Example SQL query used to identify highly engaged users:

```
SELECT `Customer ID`, `Time on Site`  
  
FROM `casestudy-490307.Ecommerce.customer_info`  
  
ORDER BY `Time on Site` DESC  
  
LIMIT 10;
```

The screenshot displays the Google Cloud BigQuery console interface. At the top, the browser address bar shows the URL: `console.cloud.google.com/bigquery?project=casestudy-490307&supportedpurview=project&ws=11m1411m313m211scasestudy-49030712sEcommerce...`. The console header includes the Google Cloud logo, the project name 'casestudy', and a search bar. Below the header, the left sidebar shows the project hierarchy: 'casestudy-490307' > 'Ecommerce' > 'customer_info'. The main area is titled 'Untitled query' and contains a SQL query: `1 SELECT 'Customer ID', 'Time on Site'`
`2 FROM 'casestudy-490307.Ecommerce.customer_info'`
`3 ORDER BY 'Time on Site' DESC`
`4 LIMIT 10;`. The query has been executed successfully, as indicated by the 'Query completed' status. Below the query, the 'Query results' section shows a table with 6 rows and 2 columns: 'Customer ID' and 'Time on Site'. The results are:

Row	Customer ID	Time on Site
1	1003	486.3
2	1006	456.2
3	1008	450.0
4	1006	400.0
5	1004	378.5
6	1010	350.0

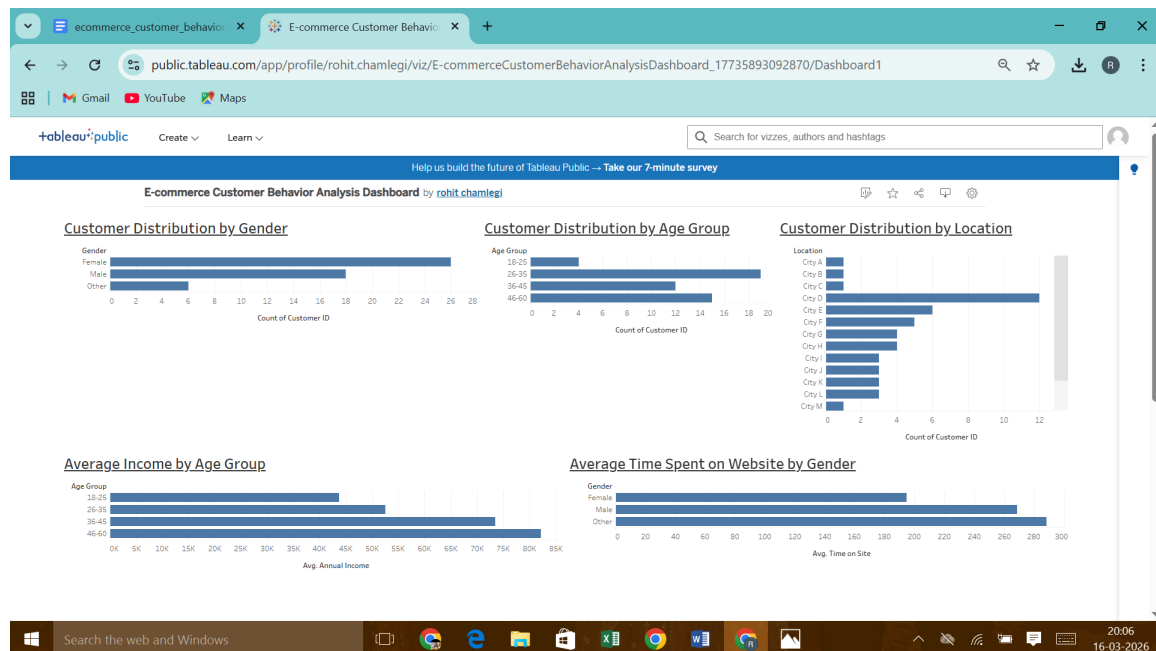
. On the right side, the 'Reference' panel shows the schema for 'customer_info', listing fields like 'Customer ID' (INTEGER), 'Age' (INTEGER), 'Gender' (STRING), 'Location' (STRING), 'Annual Income' (INTEGER), 'Purchase History' (STRING), 'Browsing History' (STRING), 'Product Reviews' (STRING), 'Time on Site' (FLOAT), and 'Age Group' (STRING). The bottom of the image shows the Windows taskbar with various application icons and the system clock displaying 20:30 on 15-03-2026.

8. Data Visualization

Data visualizations were created using Tableau to better understand customer behavior.

The dashboard includes:

- Customer Distribution by Gender
- Customer Distribution by Age Group
- Average Income by Age Group
- Average Time Spent on Website by Gender
- Customer Distribution by Location



9. Key Insights

- The 26–35 age group represents the largest customer segment.
- Higher income customers tend to spend more time on the platform.
- Engagement levels vary across gender groups.
- Certain locations contribute a larger share of customers.
- A small number of users spend significantly more time browsing the platform.

10. Business Recommendations

1. Target marketing campaigns toward the 26–35 age segment.
2. Offer personalized promotions to high-income customers.
3. Improve website engagement with personalized product recommendations.
4. Focus marketing efforts on locations with higher customer concentration.

11. Future Analysis

Future analysis could include:

- Customer segmentation using clustering
- Customer lifetime value prediction
- Predictive modeling for churn

These methods could help businesses gain deeper insights and improve decision-making.

12. Conclusion

This project demonstrates how customer demographic and behavioral data can be transformed into meaningful insights using data analytics techniques.

By using Google Sheets for data cleaning, Google BigQuery for SQL analysis, and Tableau for visualization, the project identified important patterns in customer demographics and engagement behavior.

These insights can help e-commerce businesses improve marketing strategies and better understand their customers.

13. Project Resources

Dataset Source:

Customer Behavior Dataset – Kaggle

The dataset used in this project was obtained from Kaggle.

Kaggle is a popular platform that provides open datasets for data science and analytics projects.

Link:

<https://www.kaggle.com/datasets/paulsamuelwe/e-commerce-customer-behaviour-dataset>

Data Cleaning:

https://docs.google.com/spreadsheets/d/1r9dNdfAe_PbWMDWhiFDkSut1t0a420J9/edit?usp=sharing&ouid=101217445674909205800&rtpof=true&sd=true

Interactive Dashboard:

https://public.tableau.com/app/profile/rohit.chamlegi/viz/E-commerceCustomerBehaviorAnalysisDashboard_17735893092870/Dashboard1